$$P(\lambda) = \psi^*(\lambda)\psi(\lambda).$$

8.1.2 Functions as Vectors

Up until now, the systems we have studied have had finite dimensional state-vectors. For example, the simple spin is described by a two-dimensional space of states. For this reason, the observables have had only a finite number of possible observable values. But there are more complicated observables that can have an infinite number of values. An example is a particle. The coordinates of a particle are observables, but, unlike spin, the coordinates have an infinite number of possible values. For instance, a particle moving along the x axis can be found at any real value of x. In other words, x is a continuously infinite variable. When the observables of a system are continuous, the wave function truly becomes a function of a continuous variable. To apply quantum mechanics to this kind of system, we have to expand the idea of vectors to include functions.

Functions are functions, and vectors are vectors—they seem like different things, so in what sense are functions vectors? If you think of vectors as arrows pointing in three-dimensional space, then they are not the same as functions. But if you take the broader view of vectors as a set of mathematical objects satisfying certain postulates, then functions can indeed form a vector space. Such a vector space is often called a Hilbert space after the mathematician David Hilbert.

Let's consider the set of complex functions $\psi(x)$ of a single real variable x. By complex functions, I mean that for each x, $\psi(x)$ is a complex number. On the other hand, the independent variable x is an ordinary real variable. It can take on any real value from $-\infty$ to $+\infty$.

Now, let's nail down what we mean when we say "Functions are vectors." This is not a loose analogy or a metaphor. With appropriate restrictions (that we'll come back to), functions like $\psi(x)$ *satisfy the mathematical axioms that define a vector space*. We mentioned this idea briefly in Section 1.9.2, and now we'll make full use of it. Looking back at